

33.4 Covariance models for longitudinal data

A longitudinal study follows subjects over time. Its data are clustered by subject, but with a structure the previous sections have not used: observations inside a cluster are *ordered*, and the distance between two of them matters. Compound symmetry ignores the ordering; an unstructured covariance uses it but pays $m(m+1)/2$ parameters. Between the extremes lies a catalogue of models with two to four parameters, each a different story about where the correlation comes from.

33.4.1 Three sources of correlation

A longitudinal measurement is usefully read as the sum of three parts, a decomposition due to Diggle (1988):

$$y_{ij} = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + \mathbf{z}_{(ij)}^\top \mathbf{u}_i + W_i(t_{ij}) + e_{ij}. \quad (33.4.1)$$

The vector $\mathbf{u}_i$ holds **random effects**: a subject's own level or slope, anything persisting for the whole study. The process $W_i(\cdot)$ is **serial correlation**, making nearby times alike and distant times nearly independent. And e_{ij} is **measurement error**. The three leave different fingerprints — correlation that does not decay with the gap, correlation that does, and a discontinuity at gap zero — and most models in use are [Eq. 33.4.1](#) with one or two parts dropped.

Definition 33.4.1 — Covariance models for a subject's observations

Let subject i be observed at times $t_{i1} < \dots < t_{im_i}$, and write $\boldsymbol{\Sigma}_i$ for the covariance matrix of its response vector, with entries σ_{jk} and time gap $d_{jk} = |t_{ij} - t_{ik}|$. The standard models are

Model	σ_{jk}	Parameters
independence	$\sigma^2 \delta_{jk}$	1
compound symmetry	$\sigma_u^2 + \sigma^2 \delta_{jk}$	2
first-order autoregressive	$\sigma^2 \rho^{ j-k }$	2
exponential (spatial power)	$\sigma^2 \rho^{d_{jk}}$	2
autoregressive plus random intercept	$\sigma_u^2 + \sigma^2 \rho^{ j-k }$	3
Toeplitz of order q	$\sigma^2 c_{ j-k }$, with $c_0 = 1$ and $c_u = 0$ for $u \geq q$	q
random intercept and slope	$(1, t_{ij}) \mathbf{D} (1, t_{ik})^\top + \sigma^2 \delta_{jk}$	4
unstructured	σ_{jk} , unrestricted	$m(m+1)/2$

Each model must also be nonnegative definite; the constraint is $\sigma_u^2, \sigma^2 \geq 0$ and $|\rho| < 1$ for the first five, $\mathbf{D}$ nonnegative definite for the random-coefficient model, and a genuine restriction on $(c_1, \dots, c_{q-1})$ for the Toeplitz model ([Exercise \(B2\)](#)).

Compound symmetry is Definition 33.2.1, what a random intercept alone produces: it makes

the correlation between the first and last visit equal to that between consecutive visits, which over five years it rarely is. The autoregressive model is Definition 21.6.1 within a subject and needs equally spaced occasions; the exponential model is its continuous-time version. A random intercept added to either puts a floor under the decaying correlation, and that two-source version of [Eq. 33.4.1](#) fits much real longitudinal data. The Toeplitz model lets the decay take an arbitrary shape, truncated after $q - 1$ lags. Unstructured assumes nothing, and is right when the occasions are few, common and complete and the subjects comfortably outnumber $m(m + 1)/2$: the covariance implicit in Proposition 33.3.4.

Warning. The covariance model is a choice with consequences. It changes the standard errors of the fixed effects, and — unlike in the balanced cases of the previous sections — the estimates themselves whenever the design is unbalanced or the regressors vary within subjects.

33.4.2 Choosing, and what goes wrong if you choose badly

Proposition 33.4.1 — Working covariances, likelihood comparisons and empirical standard errors

Let the N subjects be independent, with $E(\mathbf{Y}_i) = \mathbf{X}_i\boldsymbol{\beta}$ and true covariance $\boldsymbol{\Sigma}_i^0$. Let $\boldsymbol{\Sigma}_i(\boldsymbol{\theta})$ be a working covariance model whose eigenvalues are bounded above and away from zero uniformly in i and, for fixed $\boldsymbol{\theta}$, let

$$\tilde{\boldsymbol{\beta}}(\boldsymbol{\theta}) = \mathbf{A}(\boldsymbol{\theta})^{-1} \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i(\boldsymbol{\theta})^{-1} \mathbf{Y}_i, \quad \mathbf{A}(\boldsymbol{\theta}) = \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i(\boldsymbol{\theta})^{-1} \mathbf{X}_i,$$

assumed nonsingular.

- (a) $\tilde{\boldsymbol{\beta}}(\boldsymbol{\theta})$ is unbiased for every $\boldsymbol{\theta}$, whether or not the working model is correct, with exact covariance $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$ where $\mathbf{B} = \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \boldsymbol{\Sigma}_i^0 \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i$. If the working model is correct, $\mathbf{B} = \mathbf{A}$ and the covariance is $\mathbf{A}^{-1}$.
- (b) The **empirical** covariance estimate $\hat{\mathbf{U}}_E$ replaces $\boldsymbol{\Sigma}_i^0$ by $\mathbf{r}_i \mathbf{r}_i^\top$, with $\mathbf{r}_i = \mathbf{Y}_i - \mathbf{X}_i \tilde{\boldsymbol{\beta}}$. Under the conditions (K1)–(K3) of [Section 33.2](#), applied with $\boldsymbol{\Sigma}_i(\boldsymbol{\theta})^{-1} \mathbf{X}_i$ in place of $\mathbf{X}_i$, it satisfies $n(\hat{\mathbf{U}}_E - \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}) \rightarrow \mathbf{0}$ in probability as $N \rightarrow \infty$, whatever $\boldsymbol{\Sigma}_i^0$ is.
- (c) Let $\mathbf{K}$ be any $n \times (n - p)$ matrix whose columns form an orthonormal basis of $\mathcal{C}(\mathbf{X})^\perp$. The restricted log likelihood of a covariance model is the log likelihood of $\mathbf{K}^\top \mathbf{Y}$ (Definition 32.4.1) and does not depend on the choice of $\mathbf{K}$ (Theorem 32.4.2). Hence two covariance models fitted with the *same* $\mathbf{X}$ have restricted likelihoods that are densities of one and the same random vector, and their difference is a genuine log likelihood ratio. If the models differ in $\mathbf{X}$, the two are densities of different random vectors and cannot be compared.

Proof.

- (a) $\tilde{\boldsymbol{\beta}} = \mathbf{A}^{-1} \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{Y}_i$ is a fixed linear function of $\mathbf{Y}$, so $E\tilde{\boldsymbol{\beta}} = \mathbf{A}^{-1} \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \boldsymbol{\beta} = \boldsymbol{\beta}$ and, by independence of the subjects, $\text{Cov}(\tilde{\boldsymbol{\beta}}) = \mathbf{A}^{-1} \{ \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \boldsymbol{\Sigma}_i^0 \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \} \mathbf{A}^{-1}$ by Theorem 2.2.1. If $\boldsymbol{\Sigma}_i^0 = \boldsymbol{\Sigma}_i$, the middle sum telescopes to $\mathbf{A}$.

- (b) Put $\tilde{\mathbf{X}}_i = \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i$, so that $\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\sum_i \tilde{\mathbf{X}}_i^\top \mathbf{X}_i)^{-1} \sum_i \tilde{\mathbf{X}}_i^\top \boldsymbol{\varepsilon}_i$ and the empirical estimate is the same expression with $\mathbf{r}_i \mathbf{r}_i^\top$ in the middle. The cluster-sandwich argument of [Section 33.2](#) — that is, the proof of Theorem 21.4.2 with the cluster in place of the observation — goes through with $\tilde{\mathbf{X}}_g$ in the meat and $\mathbf{X}_g$ in the bread: the assumed bounds on the eigenvalues of $\boldsymbol{\Sigma}_i$ make $\|\tilde{\mathbf{X}}_i\|$ satisfy (K2) whenever $\|\mathbf{X}_i\|$ does, and (K1) and (K3) are unchanged.
- (c) The first two sentences are Definition 32.4.1 and Theorem 32.4.2, and $\mathbf{K}$ is determined up to an orthogonal rotation by $\mathcal{C}(\mathbf{X})$. Two models with the same $\mathbf{X}$ therefore attach densities to the same $(n - p)$ -dimensional random vector, so their ratio is a likelihood ratio; two models with different $\mathbf{X}$ attach them to different random vectors, and the ratio is not even dimensionless when the two $\mathbf{X}$ have different ranks. ■

Part (c) has a practical corollary: **fix the mean model first and compare covariance models by restricted likelihood; compare mean models by ordinary likelihood, or by a Wald or F test, holding the covariance fixed.** The criteria of [Chapter 29](#) then apply to the restricted likelihood as to any likelihood (Definition 29.2.2):

$$\text{AIC}_R = -2\ell_R + 2q, \quad \text{BIC}_R = -2\ell_R + q \log(n - p), \quad (33.4.2)$$

with q the number of covariance parameters and $n - p$ the number of error contrasts, the dimension of the data whose likelihood ℓ_R is.

Part (a) is why all this is less dangerous than it sounds: a wrong covariance model costs efficiency, not validity, since $\boldsymbol{\beta}$ stays unbiased and part (b) supplies a standard error that is right anyway — the longitudinal version of the sandwich argument of [Section 21.4](#), and the basis of the generalized estimating equations (Definition 40.4.1) of [Chapter 40](#).

Warning. The empirical standard error protects against the covariance model and nothing else. If $E(\mathbf{Y}_i) \neq \mathbf{X}_i \boldsymbol{\beta}$, part (a) fails at the first line and there is nothing left to be robust about; it does not protect against missing data that depend on the unobserved responses; and it needs N large in the sense of (K1)–(K3), being worse than the model-based estimate with a dozen subjects.

33.4.3 Checking the fit of a covariance model

Two displays do most of the work, both built from residuals $\hat{r}_{ij} = y_{ij} - \mathbf{x}_{(ij)}^\top \hat{\boldsymbol{\beta}}$ after a fit of the mean model. The **empirical variogram** plots, against the time gap u , the average of $\frac{1}{2}(\hat{r}_{ij} - \hat{r}_{ik})^2$ over same-subject pairs at that gap. Its population version $\gamma(u) = \frac{1}{2} \text{Var}(r_{ij} - r_{ik})$ is, for [Eq. 33.4.1](#), $\sigma_e^2 + \sigma_s^2\{1 - \rho(u)\}$, rising from σ_e^2 to $\sigma_e^2 + \sigma_s^2$; the gap between that plateau and the total residual variance $\sigma_u^2 + \sigma_s^2 + \sigma_e^2$ measures σ_u^2 , the part of the correlation that never decays. The **residual correlogram** plots the average correlation of residuals at gap u : compound symmetry predicts a horizontal line, the autoregressive model a geometric decay to zero, the combination a decay to a positive floor. Comparing empirical with fitted curves shows quickly which model does violence to the data.

Example. (Investment in eleven firms)

Grunfeld's panel records gross investment, market value and capital stock for 11 US firms in each of the 20 years from 1935 to 1954, a complete rectangle with $n = 220$. Take log investment as the response and the mean model $E(y) = \beta_0 + \beta_1 t + \beta_2 \log(\text{value})$, with t in decades from the middle of the record, so $p = 3$. An unstructured covariance would need 210 parameters from 11 subjects, which is hopeless. Fitting the alternatives by restricted maximum likelihood gives

Covariance model	q	$-2\ell_R$	AIC_R	BIC_R
independence	1	476.95	478.95	482.33
compound symmetry	2	114.83	118.83	125.59
autoregressive	2	49.89	53.89	60.65
autoregressive + random intercept	3	29.87	35.87	46.01
random intercept and slope	4	98.66	106.66	120.18
Toeplitz of order 4	4	147.97	155.97	169.49

Independence is hopeless: the firms differ enormously in size, and log investment is nearly constant within a firm relative to the spread between firms. Compound symmetry captures that and improves $-2\ell_R$ by 362, the autoregressive model on the same two parameters does better still, and the combination is best on both criteria, by 82.96 of AIC_R over compound symmetry. The random-coefficient model, which gives each firm its own trend, describes these data worse than a persistent level plus decaying serial correlation.

Figure 33.4.1 shows why. The residual correlations are 0.953 at one year, 0.918 at two, 0.884 at five and 0.839 at ten: high everywhere, because the firm's level dominates, but visibly decaying, which compound symmetry cannot reproduce. The variogram rises to a plateau of about 0.099, far below the total residual variance of 0.4839; the gap is the between-firm variance. The selected model is not innocent either: its fitted correlogram flattens onto its own floor after three or four years while the data go on decaying. That is what the diagnostic is for, and a reason to read the empirical standard error of Proposition 33.4.1(b) alongside the model-based one.

With the selected model the coefficient of $\log(\text{value})$ is 0.6295, with a model-based standard error of 0.0712 and an empirical one of 0.0608, and the coefficient of t is 0.4261 with standard errors 0.0600 and 0.0601. Ordinary least squares, treating the 220 observations as independent, reports 0.9833 and 0.3768 with standard errors 0.0351 and 0.0821. The two disagree substantially about the elasticity with respect to market value, because least squares reads it mostly from differences between firms while generalized least squares, having absorbed the firm levels into the covariance, reads it from movements within firms — the between-versus-within distinction of Section 6.6 again, and a question about the subject matter.

```

import numpy as np
import statsmodels.api as sm

df = sm.datasets.grunfeld.load_pandas().data
firms = sorted(df["firm"].unique())
years = np.sort(df["year"].unique())
N, m = len(firms), len(years)                                # 11 firms, 20 years
t = (years - years.mean()) / 10                               # centred and scaled time

Ylist, Xlist = [], []
for f in firms:
    d = df[df["firm"] == f].sort_values("year")
    Ylist.append(np.log(d["invest"].to_numpy()))
    Xlist.append(np.column_stack([np.ones(m), t, np.log(d["value"].to_numpy())]))
p = Xlist[0].shape[1]
n = N * m

```

[Run this code online](#)

With a common Σ , $-2\ell_R$ is $N \log|\Sigma| + \sum_i \mathbf{r}_i^\top \Sigma^{-1} \mathbf{r}_i + \log|\sum_i \mathbf{X}_i^\top \Sigma^{-1} \mathbf{X}_i|$ plus a constant, evaluated at the generalized least squares $\hat{\beta}$, the third term being what distinguishes it from the ordinary likelihood (Theorem 32.4.2).

```

from scipy.optimize import minimize

def reml(Sigma, Ylist, Xlist):
    """-2 x the restricted log likelihood, and the GLS estimate, for a common Sigma.
    """
    L = np.linalg.cholesky(Sigma)
    logdet = 2 * np.sum(np.log(np.diag(L)))
    solve = lambda B: np.linalg.solve(Sigma, B)
    XtWX = sum(X.T @ solve(X) for X in Xlist)
    XtWy = sum(X.T @ solve(y) for X, y in zip(Xlist, Ylist))
    beta = np.linalg.solve(XtWX, XtWy)
    quad = sum((y - X @ beta) @ solve(y - X @ beta) for X, y in zip(Xlist, Ylist))
    s, ld_info = np.linalg.slogdet(XtWX)
    n, p = len(Ylist) * Sigma.shape[0], len(beta)
    return len(Ylist) * logdet + quad + ld_info + (n - p) * np.log(2 * np.pi), beta

lag = np.abs(np.subtract.outer(np.arange(m), np.arange(m)))
Zrc = np.column_stack([np.ones(m), t])                                # random intercept and slope
basis

def build(name, th):
    """Sigma for each covariance model, from an unconstrained parameter vector."""
    if name == "independence":
        return np.exp(th[0]) * np.eye(m)
    if name == "compound symmetry":
        return np.exp(th[0]) * np.eye(m) + np.exp(th[1]) * np.ones((m, m))
    if name == "AR(1)":
        return np.exp(th[0]) * np.tanh(th[1])**lag
    if name == "AR(1) + random intercept":
        return (np.exp(th[0]) * np.tanh(th[1])**lag + np.exp(th[2]) * np.ones((m, m)
    ))
    if name == "random intercept and slope":
        D = np.array([[np.exp(th[1]), th[3]], [th[3], np.exp(th[2])]])
        return np.exp(th[0]) * np.eye(m) + Zrc @ D @ Zrc.T

```

```

if name == "Toeplitz(4)":
    a = np.concatenate([[1.0], th[1:4]])          # a moving average of order 3
    ...
    g = np.array([a[:4 - u] @ a[u:] for u in range(4)])
    c = np.concatenate([g / g[0], np.zeros(m - 4)]) # ... so the band is
nonneg. definite
    return np.exp(th[0]) * c[lag]
raise ValueError(name)

MODELS = {"independence": 1, "compound symmetry": 2, "AR(1)": 2,
          "AR(1) + random intercept": 3, "random intercept and slope": 4,
          "Toeplitz(4)": 4}

def fit(name, start):
    def obj(th):
        S = build(name, th)
        if np.min(np.linalg.eigvalsh(S)) <= 1e-9:
            return 1e6
        return reml(S, Ylist, Xlist)[0]
    best = min((minimize(obj, s, method="Nelder-Mead",
                        options={"maxiter": 20000, "xatol": 1e-9, "fatol": 1e-9})
               for s in start), key=lambda r: r.fun)
    S = build(name, best.x)
    d2l, beta = reml(S, Ylist, Xlist)
    q = MODELS[name]
    return {"m2l1": d2l, "aic": d2l + 2 * q, "bic": d2l + q * np.log(n - p),
           "q": q, "Sigma": S, "beta": beta}

starts = {"independence": [[-1.0]],
          "compound symmetry": [[-1.0, -1.0]],
          "AR(1)": [[-1.0, 1.0]],
          "AR(1) + random intercept": [[-1.5, 1.0, -1.5]],
          "random intercept and slope": [[-2.0, -1.0, -2.0, 0.0], [-3.0, -2.0, -3.0,
0.0]],
          "Toeplitz(4)": [[-1.0, 1.5, 1.0, 0.5], [-1.0, 2.5, 2.0, 1.0]]}
res = {name: fit(name, starts[name]) for name in MODELS}
for name in MODELS:
    r = res[name]
    print(f"{name:28s} q={r['q']} -2logL_R={r['m2l1']:9.3f} "
          f"AIC={r['aic']:9.3f} BIC={r['bic']:9.3f}")

```

[Run this code online](#)

```

best_name = min(MODELS, key=lambda k: res[k]["aic"])
Xall = np.vstack(Xlist)
yall = np.concatenate(Ylist)
Sbest = res[best_name]["Sigma"]
A = np.linalg.inv(sum(X.T @ np.linalg.solve(Sbest, X) for X in Xlist))
beta = res[best_name]["beta"]
resid = [y - X @ beta for X, y in zip(Xlist, Ylist)]
meat = sum(np.outer(X.T @ np.linalg.solve(Sbest, r), X.T @ np.linalg.solve(Sbest, r)
)
           for X, r in zip(Xlist, resid))
cov_robust = A @ meat @ A * N / (N - 1)          # empirical ("sandwich")
covariance
XtXinv = np.linalg.inv(Xall.T @ Xall)
beta_ols = XtXinv @ Xall.T @ yall
s2_ols = np.sum((yall - Xall @ beta_ols)**2) / (n - p)

```

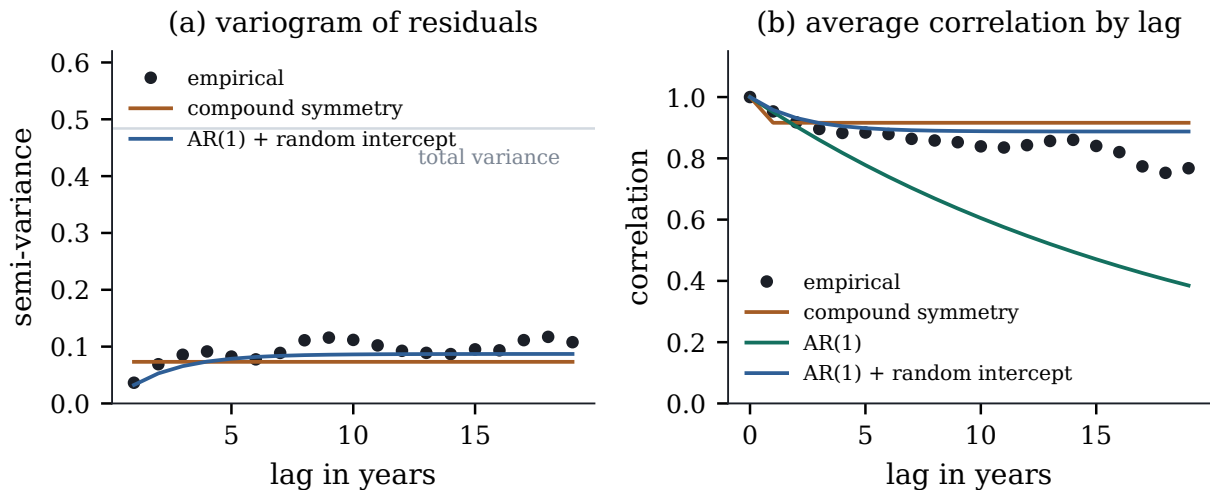


Figure 33.4.1: Checking the covariance model for the investment panel. (a) The empirical variogram of the residuals, with those implied by two fitted models; the gap between the grey total-variance line and the plateau is the between-firm variance. (b) The average residual correlation at each gap, with three fitted models.

```
for j, lab in enumerate(["intercept", "time/10", "log value"]):
    print(f"{lab:10s} GLS {beta[j]: .4f} model se {np.sqrt(A[j, j]):.4f} "
          f"empirical se {np.sqrt(cov_robust[j, j]):.4f} "
          f"| OLS {beta_ols[j]: .4f} se {np.sqrt(s2_ols * XtXinv[j, j]):.4f}")
```

[Run this code online](#)

33.4.4 Missing data

Longitudinal data are rarely complete. The mixed-model machinery handles ragged data without complaint — [Proposition 33.4.1](#) never assumed a common m_i — an advantage over the classical analysis of [Section 33.3](#), which needs a complete rectangle. But “handles” means “computes”. If the probability of a missing value depends only on the covariates and the responses that *were* observed, the observed-data likelihood is the right one: the data are missing at random. If it depends on the value that would have been observed — dropout because a patient is not improving — the likelihood is wrong and no covariance model repairs it.

Warning. Software that fits a mixed model to incomplete longitudinal data makes a missing-at-random assumption on the analyst’s behalf, silently, and the empirical standard error of [Proposition 33.4.1\(b\)](#) does *not* protect against its failure. [Chapter 41](#) treats missing data properly.

33.4.5 Exercises

33.4.5.1 A. Check your understanding

Exercise A1.

A trial measures 60 patients at 8 visits. How many parameters does each model of

Definition 33.4.1 need? Which could not be fitted with only 10 patients, and why?

Exercise A2.

Sketch the population variogram $\gamma(u)$ for compound symmetry, for the autoregressive model and for their combination, marking the total variance on each. Which feature identifies σ_u^2 , and which identifies measurement error?

Solution. Compound symmetry gives $\gamma(u) = \sigma^2$ for every $u > 0$, flat and well below the total variance $\sigma^2 + \sigma_u^2$. The autoregressive model gives $\gamma(u) = \sigma^2(1 - \rho^u)$, rising from 0 to its total variance σ^2 . The combination rises from 0 to σ^2 and stops, leaving the gap σ_u^2 to the total. The gap between plateau and total variance is σ_u^2 ; the intercept of the variogram as $u \rightarrow 0$ is the measurement-error variance.

33.4.5.2 B. Practice

Exercise B1.

Two analysts fit the investment data of [Example \(Investment in eleven firms\)](#). The first uses the mean model of the example with an autoregressive covariance; the second adds a quadratic term in t and uses compound symmetry. They compare AIC_R . Explain what is wrong, and describe a correct sequence of comparisons.

Solution. By [Proposition 33.4.1\(c\)](#) the two restricted likelihoods are densities of different random vectors, of dimensions $n - 3$ and $n - 4$, so their difference is not a log likelihood ratio and AIC_R cannot compare them. Correct sequence: fix a rich mean model and compare covariance models by AIC_R ; then, with the chosen covariance fixed, compare mean models by a Wald or F test, or by AIC from the *full* likelihood; then refit the covariance.

Exercise B2.

Show that the Toeplitz model of [Definition 33.4.1](#) with $q = 3$ is nonnegative definite for every m iff $(1, c_1, c_2, 0, \dots)$ is the autocorrelation sequence of a moving-average process of order two, and explain why the parameterization $c_u \propto \sum_l a_l a_{l+u}$ used in the listing above satisfies this automatically.

33.4.5.3 C. Going deeper

Exercise C1.

The autoregressive model is nested in the autoregressive-plus-random-intercept model at $\sigma_u^2 = 0$, a boundary of the parameter space. Explain why the likelihood ratio statistic comparing the two does not have a $\chi^2(1)$ null distribution, state what it does have ([Proposition 32.5.1](#)), and say what this means for reading AIC_R differences as informal evidence.